

EOC240.02 – INVENT A COMMUNICATION SYSTEM FOR SPACE

By:





Teaching Points

- **TP1 Explain That Astronauts Use Radio Communication While in Space to Communicate With Other Astronauts and Ground Control**
- **TP2 Invent a Communication System for Space**
- **TP3 Discuss Communication System Inventions With the Entire Group**



- While in space, astronauts use **Ultra High Frequency (UHF) radio transceivers** to communicate with their colleagues inside the shuttle cabin
- The astronauts also sometimes use their UHF radios to **talk with ground control** during launch or landing
- The Communications Carrier Assembly (CCA) is a fabric cap worn by astronauts
- It contains **microphones and speakers** for use with the radio
- It allows for **hands-free radio communication** while wearing a spacesuit



TP1 Explain That Astronauts Use Radio Communication While in Space to Communicate With Other Astronauts and Ground Control Pt 2

- It is virtually impossible to provide regulations and procedures applicable to all possible situations associated with two-way radio communication failure
- During **two-way radio communication failure**, when confronted with a situation not covered in the regulations, **astronauts are expected to exercise good judgement in whatever alternative communication method they choose to communicate with each other**

- Q1. While in space, what do shuttle astronauts use to communicate with their colleagues inside the shuttle cabin?
- Q2. What allows hands-free radio communication while wearing a spacesuit?
- Q3. What must astronauts do during two-way radio communication failure?
- A1. While in space, shuttle astronauts use UHF radio transceivers to communicate with their colleagues inside the shuttle cabin.
- A2. The CCA is a fabric cap worn by the astronaut. It contains microphones and speakers for use with the radio.
- A3. Astronauts are expected to exercise good judgement in whatever alternative communication method they choose to communicate with each other.

- 1. Divide the cadets into groups of no more than five**
- 2. Provide the groups with a message cue card located at Annex A to focus on what they must be able to communicate using their communication invention**
- 3. Groups will have 30 minutes to invent a communication system for space**
- 4. One cadet from each group will present their group's invention to the class by communicating the message from their cue card using the invention**
- 5. Once the cadet has communicated the message using their invention, the class must determine the message. The cadet cannot communicate the message to the class using speech**
- 6. The cadet will read the message aloud to the class once they determine the message**

MESSAGE CUE CARDS

I am out of air.

I am sick.

I am stuck.

I need help.

Come here.

I am lost.

